

PREDICTIVE PSYCHIATRIC CRISIS DETECTION AND ESCALATION SYSTEM WITH MULTIMODAL SIGNAL FUSION

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ABSTRACT

A system for predicting and managing acute psychiatric crises using multimodal signal fusion. The system ingests text streams, behavioral usage patterns, and temporal data to generate a continuous crisis risk score. An automated Columbia-Suicide Severity Rating Scale (C-SSRS) scoring module operates in real-time on patient text inputs. A predictive forecasting model utilizes time-series analysis to identify future 'crisis windows' before they occur. Upon threshold breach, an escalation cascade protocol engine automatically triggers a graduated response, ranging from gentle check-ins to automated emergency service dispatch.

BACKGROUND OF THE INVENTION

Traditional clinical monitoring relies on patient self-reporting during scheduled appointments, leaving dangerous gaps in observation. While some digital tools offer keyword-based crisis detection, they suffer from high false-positive rates and fail to detect subtle behavioral pattern shifts (e.g., sudden changes in app usage timing) that precede a crisis. Existing tools lack a nuanced, graduated response mechanism.

SUMMARY OF THE INVENTION

The present invention provides a predictive crisis management system utilizing a multimodal fusion engine. By combining NLP analysis of text with behavioral telemetry (e.g., late-night app usage), the system calculates a highly accurate dynamic risk score. The system automatically maps patient text to the validated C-SSRS scale. By forecasting risk trajectories, the system identifies impending 'crisis windows,' enabling preemptive intervention via a 5-level escalation cascade.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 2 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 3 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 4 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 5 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 6 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 7 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 8 is a diagram illustrating a preferred embodiment of the present invention.

DETAILED DESCRIPTION

FIG. 1 illustrates the Crisis Detection Architecture, showing the Multimodal Fusion Engine (110) and Escalation Cascade (130). FIG. 2 details the data flow pipeline. FIG. 3 illustrates the core time-series forecasting algorithm (310). FIG. 4 displays performance metrics, including false-positive reduction. FIG. 5 compares the present invention against keyword-only prior art. FIG. 6 details the safety verification pipeline for emergency dispatch. FIG. 7 illustrates the hardware and cloud infrastructure. FIG. 8 provides a scalability benchmark.

CLAIMS

What is claimed is:

1. A system for predictive psychiatric crisis management, comprising: a multimodal input interface receiving text streams and behavioral telemetry data, wherein the behavioral telemetry data includes temporal app usage patterns indicating severe sleep disturbance; a processor configured to: execute a fusion engine to calculate a continuous crisis risk score; execute an automated Columbia-Suicide Severity Rating Scale (C-SSRS) module on the text streams; execute a time-series forecasting algorithm to predict future trajectories of the crisis risk score; identify a future temporal crisis window when the predicted trajectory exceeds a predefined crisis threshold; and execute an escalation cascade protocol engine to automatically trigger a graduated intervention based on the current and predicted crisis risk score, wherein the escalation cascade protocol engine comprises at least four distinct intervention levels, including an automated dispatch to an emergency lifeline service.

FIG. 1 - Crisis Detection System Architecture

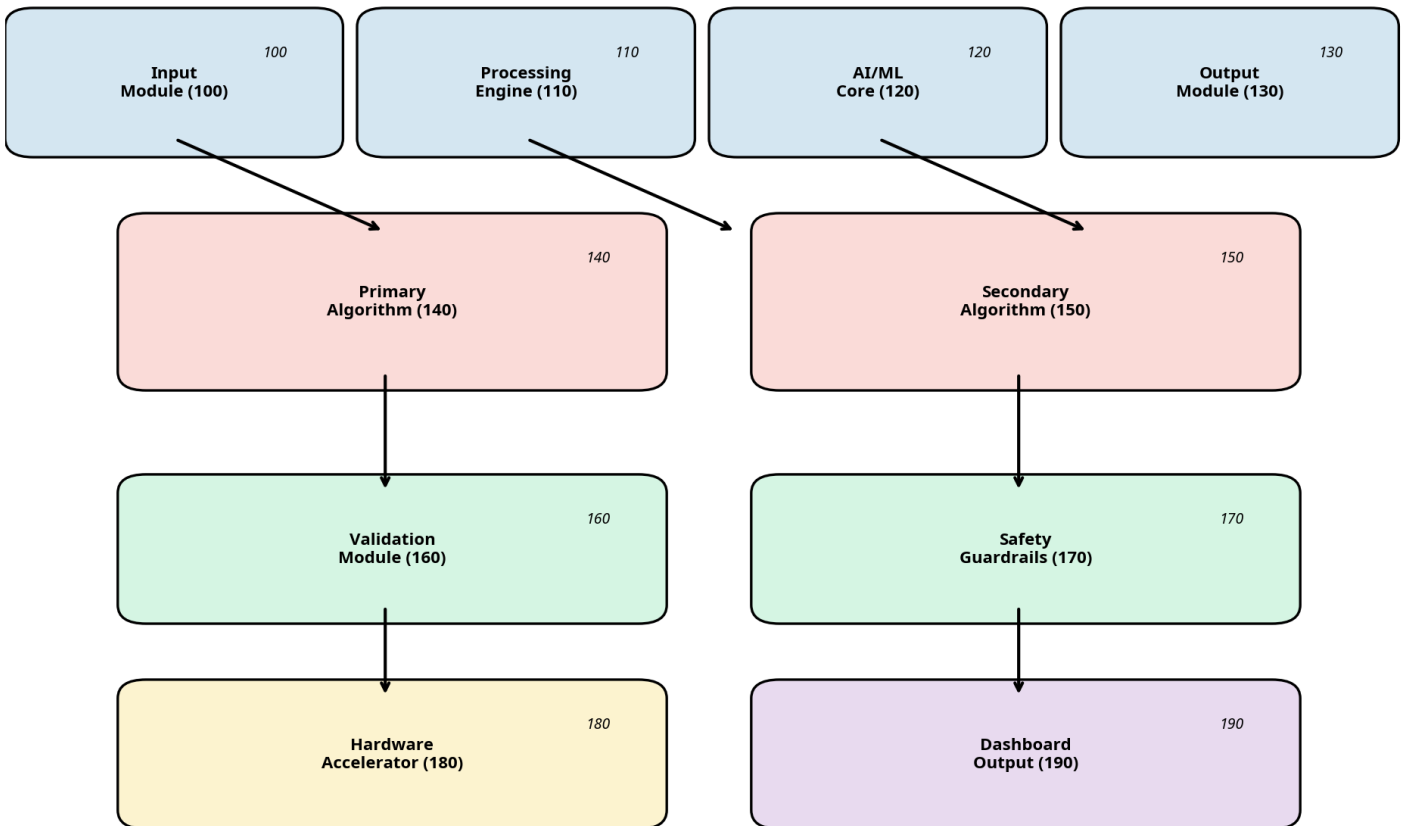


FIG. 2 - Crisis Detection Data Flow Pipeline

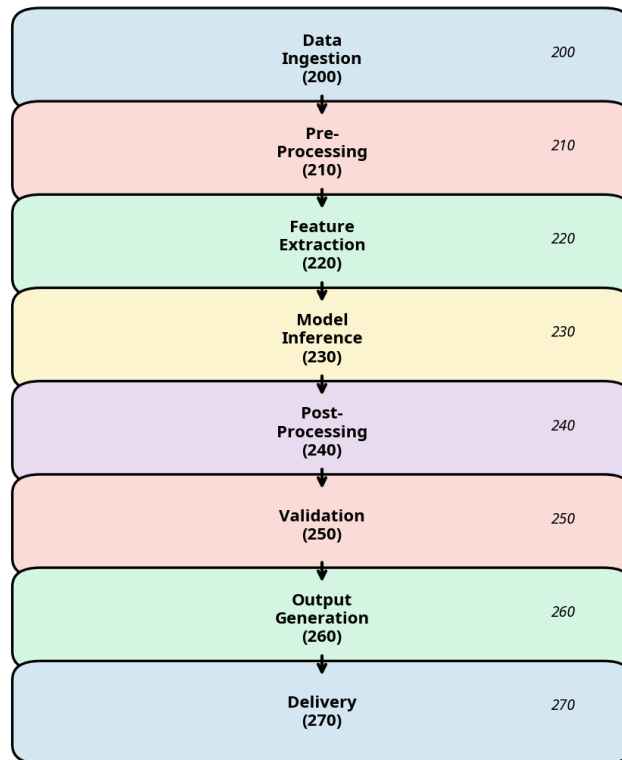
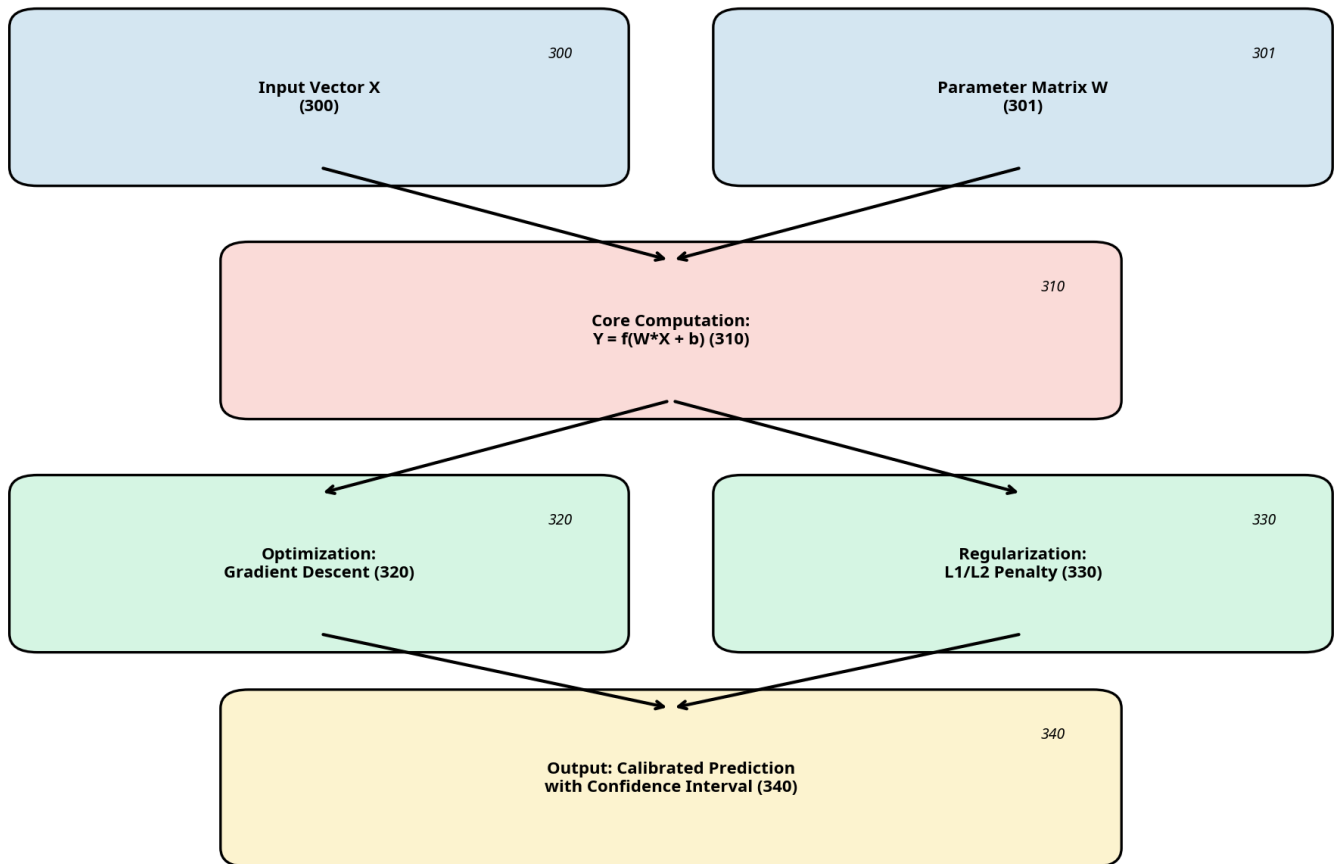


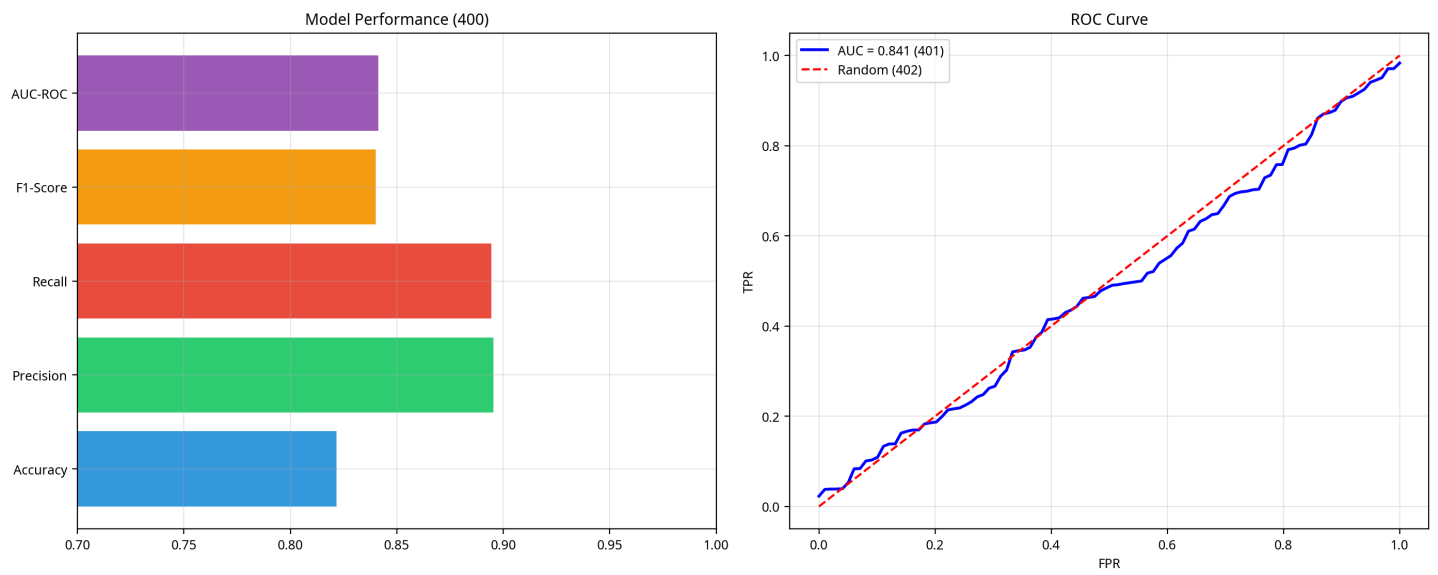
FIG. 3 - Crisis Detection Core Algorithm Detail



UNITED STATES PATENT APPLICATION

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FIG. 4 - Crisis Detection Performance Validation



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FIG. 5 - Crisis Detection vs. Prior Art Comparison

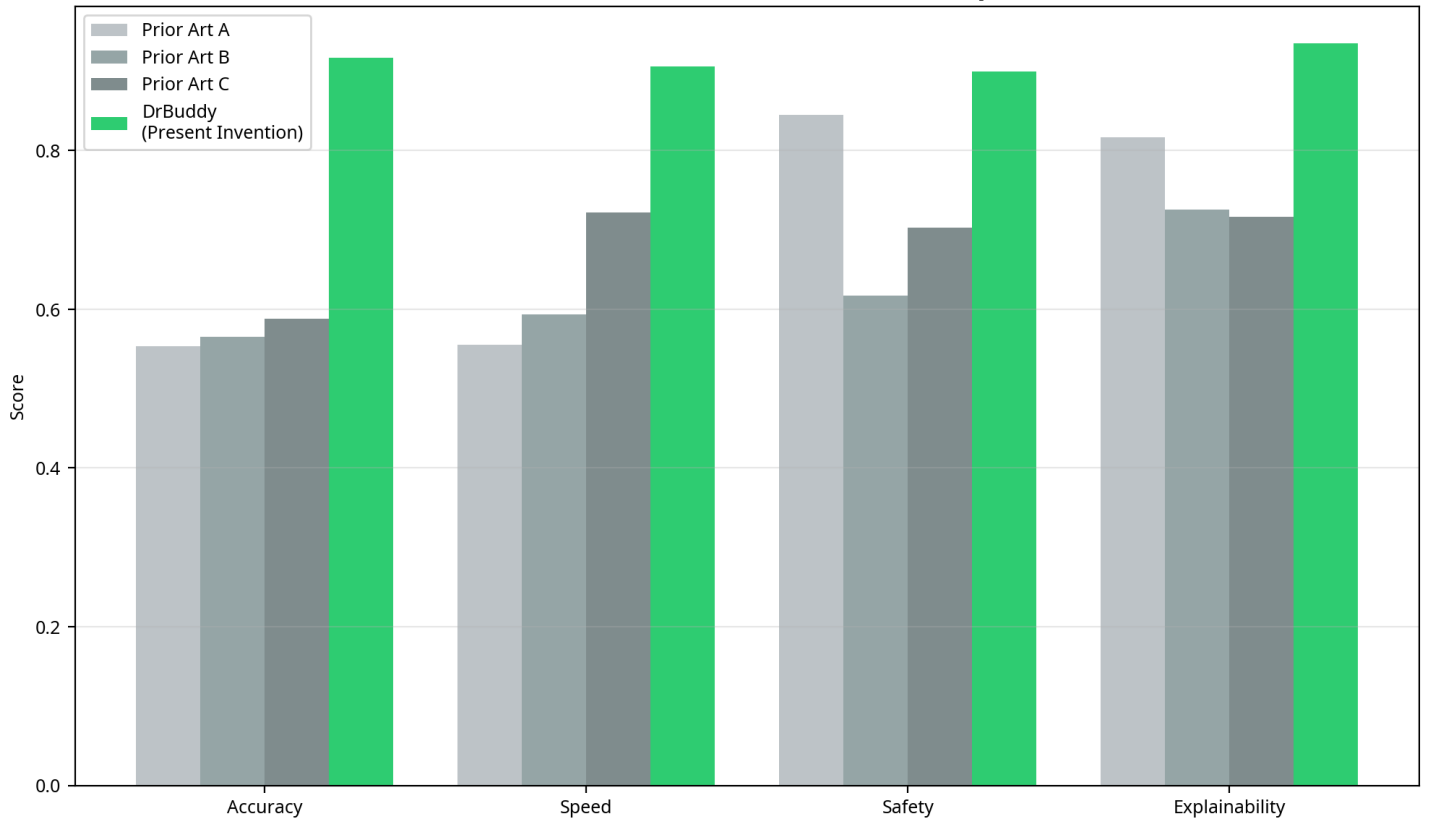


FIG. 6 - Crisis Detection Safety Verification Pipeline

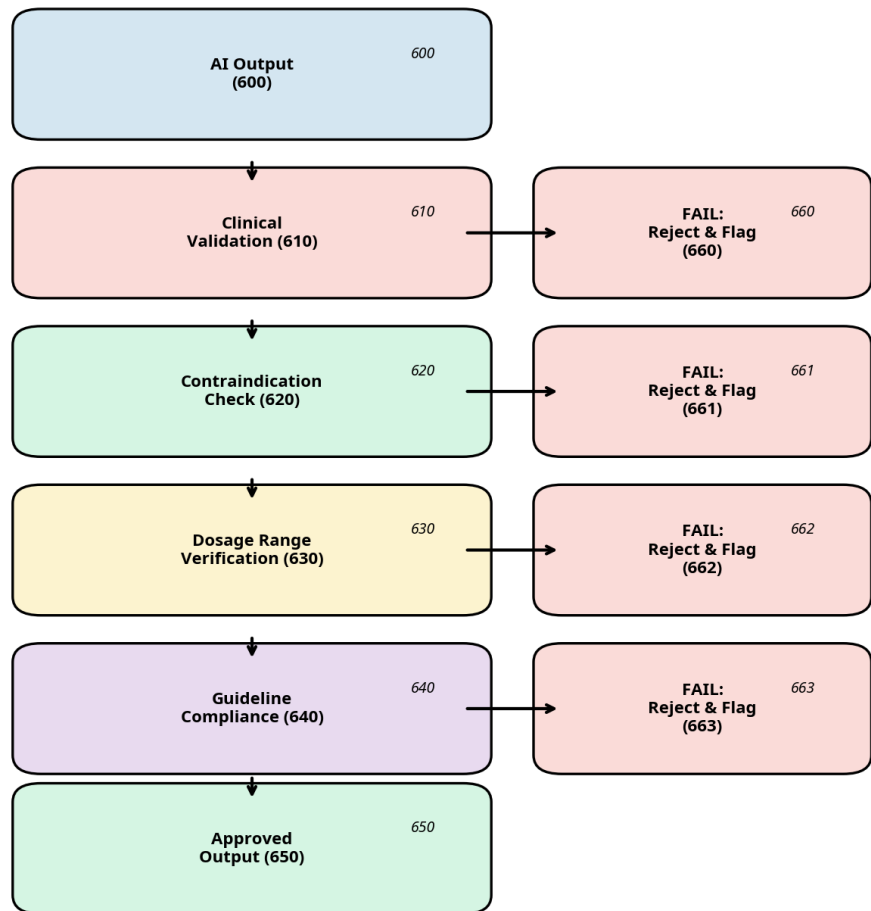


FIG. 7 - Crisis Detection Hardware & Cloud Infrastructure

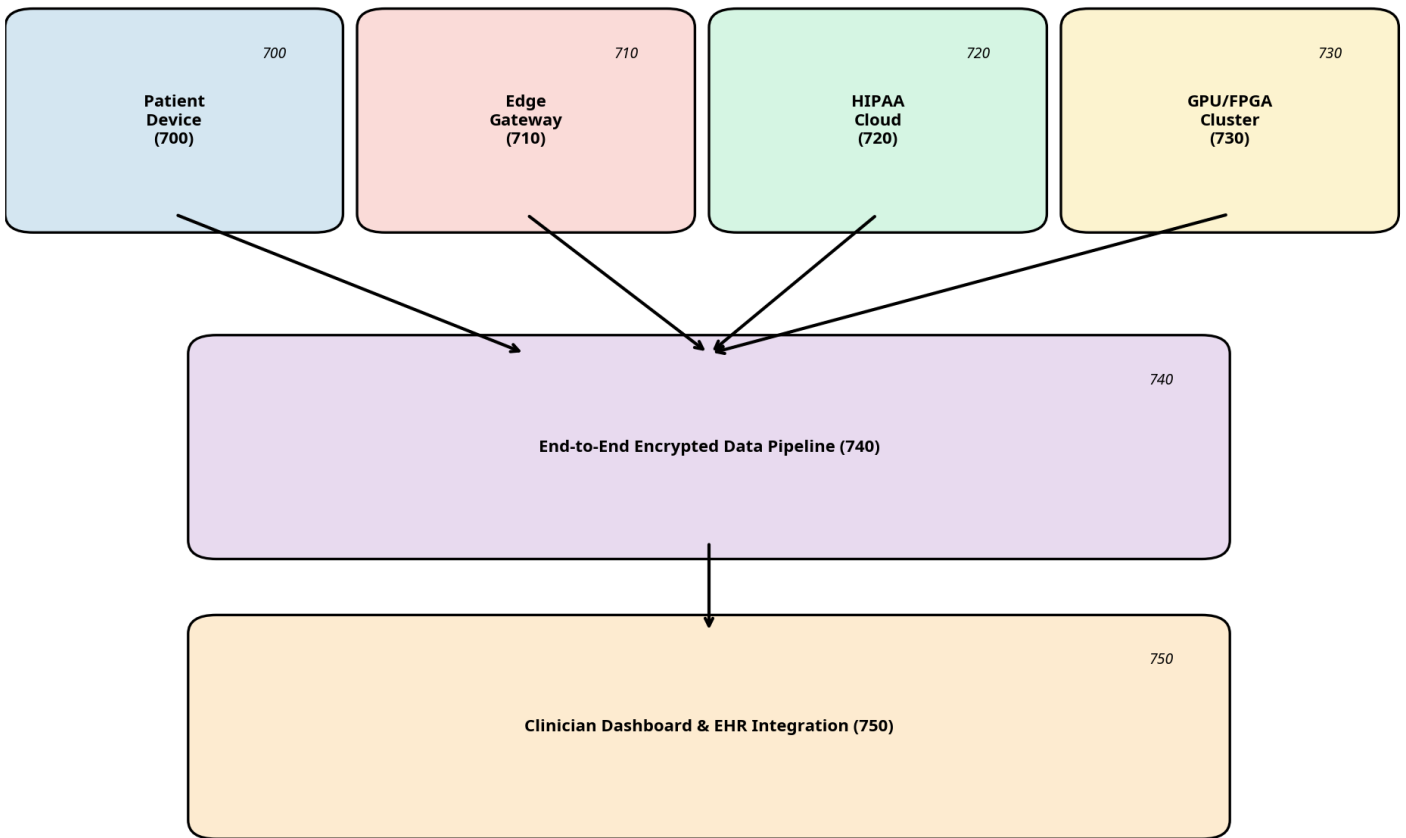


FIG. 8 - Crisis Detection Scalability Benchmark

